Financial Fraud Detection & Risk Scoring System

AI-Driven Financial Transaction Monitoring and Fraud Detection

---

Zidio Development – Data Science Internship 2026

Project 2

Developed By – Team 20

Rushda

Gauri Chavanke

Harshit Yadav

Akshat Singh Jhala

Project Duration

25th June – 25th July 2026

Technologies Used

Python • Pandas • NumPy • Scikit-learn • XGBoost • Optuna • SHAP • Kafka • Streamlit

Key Areas

Fraud Detection • Machine Learning • Anomaly Detection • Risk Scoring • Explainable AI • Real-Time Monitoring

1. INTRODUCTION

The rapid adoption of digital banking, online payments, credit cards and electronic financial services has significantly increased the volume of financial transactions. While digital transactions provide convenience and faster services, they have also created new opportunities for fraudulent activities. Financial fraud can result in direct monetary losses, compromise customer information and reduce confidence in digital financial systems.

Fraud detection is a challenging machine learning problem because fraudulent transactions generally constitute only a small fraction of the total transaction volume. In such highly imbalanced datasets, a model can achieve high accuracy simply by predicting most transactions as legitimate while failing to identify actual fraudulent transactions. Therefore, an effective fraud detection system must focus not only on overall accuracy but also on precision, recall and F1-score.

2. PROBLEM STATEMENT

The major challenge in financial fraud detection is to distinguish fraudulent transactions from legitimate transactions accurately while dealing with highly imbalanced data.

In the dataset used for this project, legitimate transactions were significantly higher than fraudulent transactions. Before balancing, the training dataset contained **39,664 non-fraudulent transactions and only 698 fraudulent transactions**. This imbalance makes conventional accuracy-based evaluation unreliable.

Another challenge is that fraudulent behaviour is not always consistent. Fraudulent transactions may differ from legitimate transactions in terms of transaction amount, transaction frequency, location, device, payment channel, failed payment attempts or other behavioural characteristics.

Therefore, the project aims to develop a system that can learn complex transaction patterns, identify fraudulent transactions, detect unusual behaviour and provide a meaningful risk assessment for individual transactions.

3. OBJECTIVES

The project was developed with the following objectives:

- To integrate multiple financial fraud datasets into a unified dataset.
- To clean and prepare transaction data for machine learning.
- To analyse transaction and behavioural patterns associated with fraud.
- To address the severe class imbalance present in the dataset.
- To develop multiple machine learning models for fraud classification.
- To compare models using appropriate fraud detection metrics.
- To optimize model performance through hyperparameter tuning.
- To explore anomaly detection for previously unseen suspicious behaviour.
- To develop a hybrid fraud risk-scoring approach.
- To provide interpretable model predictions using SHAP.
- To design a real-time fraud detection workflow using Kafka.
- To provide an interactive monitoring interface using Stream lit.

4. DATASET DESCRIPTION

The project uses four financial fraud-related datasets:

1. **Fraud Detection Dataset**
2. **Credit Card Fraud Dataset**
3. **Financial Fraud Detection Dataset**
4. **Synthetic Fraud Dataset**

The datasets were first examined individually to understand their structure, attributes and fraud distributions. Since the datasets originated from different sources, their schemas and data types were compared before integration.

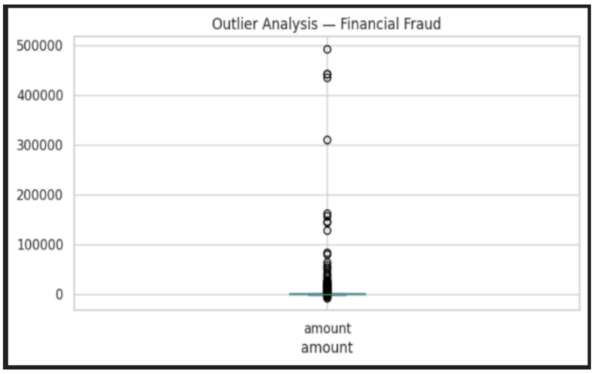
5. DATA PREPROCESSING

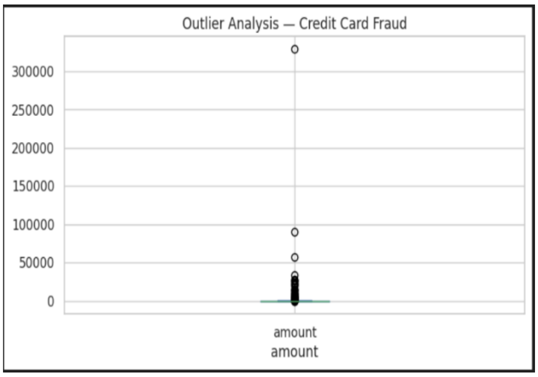
Data preprocessing was performed to convert the raw integrated data into a form suitable for statistical analysis and machine learning.

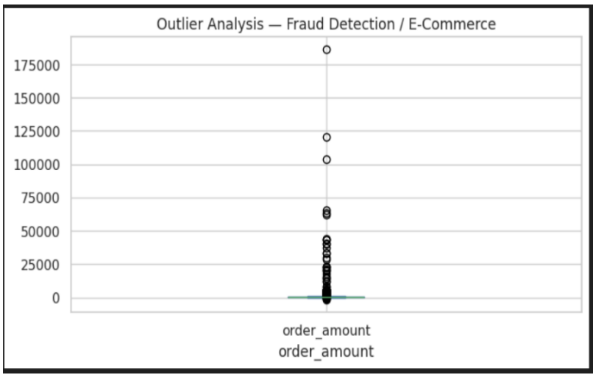
The preprocessing stage included data-type verification, missing-value analysis, duplicate checking and preparation of numerical and categorical attributes. Transaction date information was processed to derive temporal characteristics such as transaction hour, transaction day and transaction month.

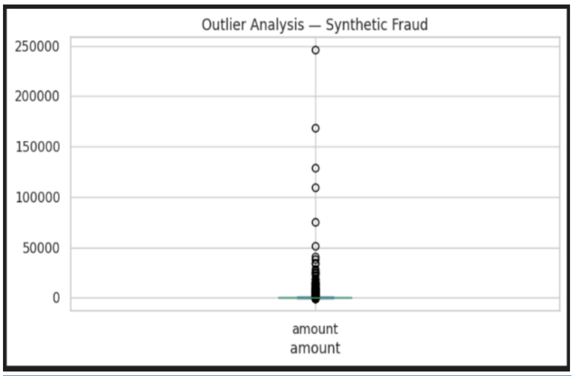
Several behavioral variables were also prepared because fraud can be associated with unusual transaction frequency, repeated failed payment attempts, unfamiliar devices or changes in transaction location.

The preprocessing stage ensured that the resulting dataset had a consistent structure and could be used reliably for subsequent exploratory analysis and model development.









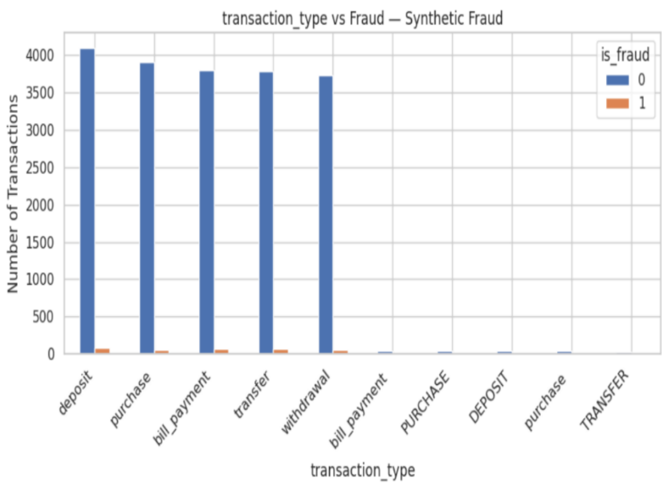
6. EXPLORATORY DATA ANALYSIS

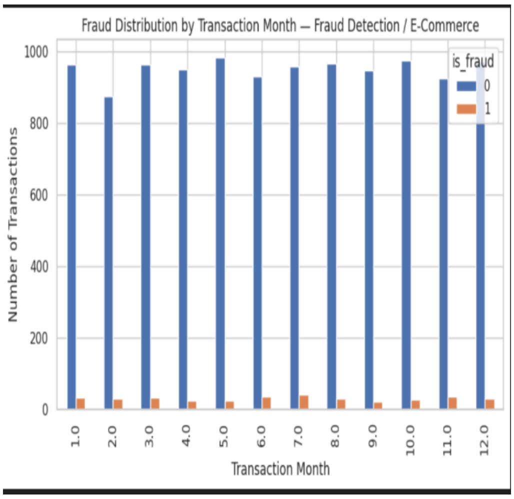
Exploratory Data Analysis was performed to understand the characteristics of fraudulent and legitimate transactions before applying machine learning algorithms.

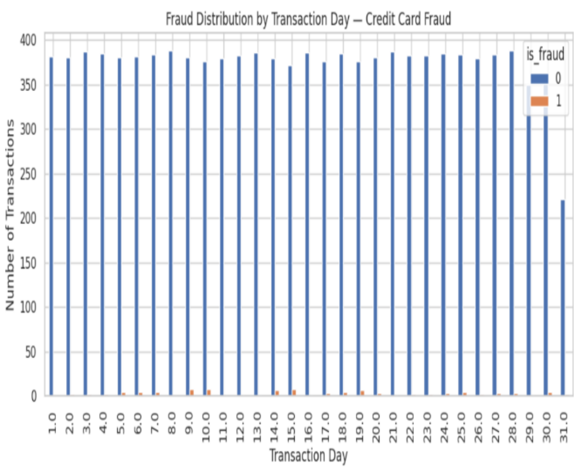
The analysis examined the distribution of the target variable, transaction amounts, transaction channels, risk scores, transaction timing and behavioral attributes. Particular attention was given to identifying characteristics that differed between fraudulent and non-fraudulent transactions.

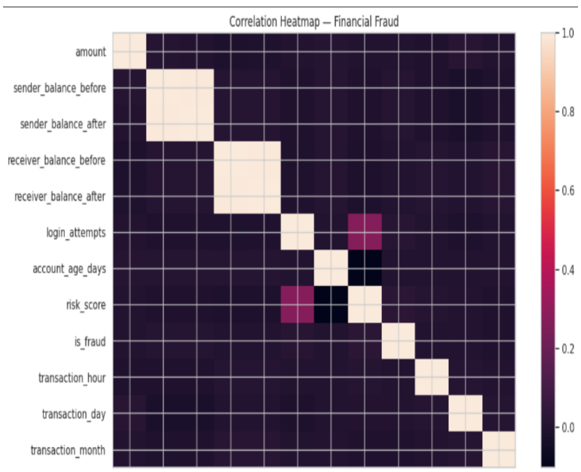
The target distribution confirmed a significant imbalance between the two classes. Additional analysis was performed on transaction frequency, device usage, geographic behaviour and failed payment attempts to identify potential indicators of suspicious activity.

The EDA stage helped establish an understanding of the dataset and guided the selection and engineering of features for the subsequent modelling stage.









7. DATA SPLITTING AND CLASS IMBALANCE

To evaluate the models on unseen data, the dataset was divided into training, validation and testing subsets.

The training data contained 39,664 legitimate transactions and only 698 fraudulent transactions. Such a distribution creates a strong bias toward the majority class and can negatively affect the ability of machine learning models to recognize fraud.

To address this issue, **SMOTE (Synthetic Minority Over-sampling Technique)** was applied only to the training data. SMOTE generated synthetic minority-class samples to increase the representation of fraudulent transactions.

After balancing, the training dataset contained **79,328 records**, consisting of **39,664 legitimate and 39,664 fraudulent transactions**.

8. MACHINE LEARNING METHODOLOGY

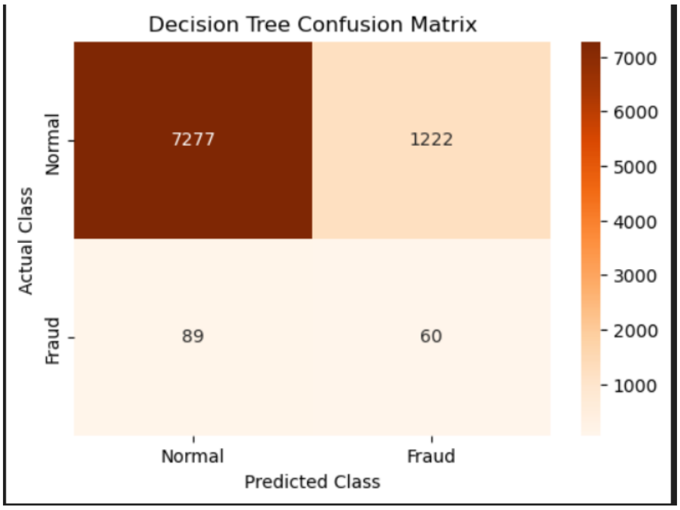
Multiple supervised learning algorithms were implemented to determine which approach was most suitable for the fraud detection problem.

Logistic Regression

Logistic Regression was selected as the baseline model because of its simplicity and interpretability. It provides a reference against which more complex machine learning models can be evaluated.

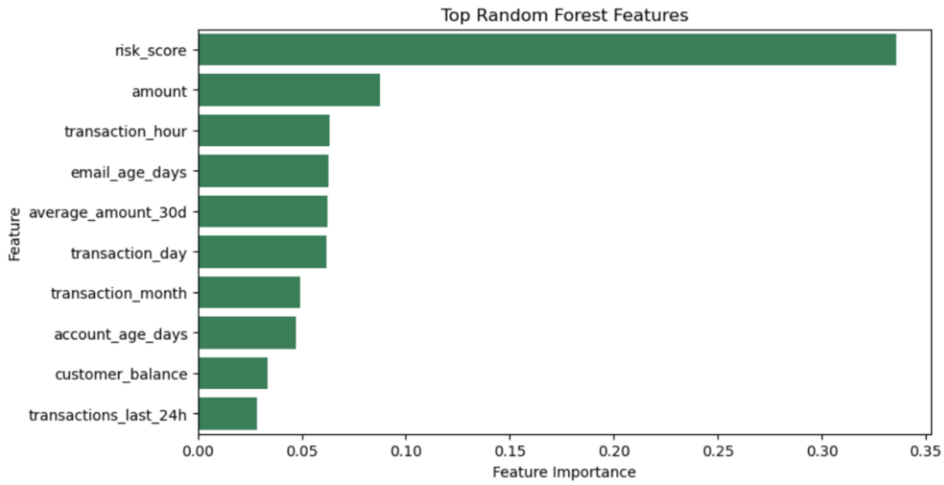
Decision Tree

Decision Tree was used to capture non-linear relationships between transaction attributes and fraud outcomes. The model makes predictions using a sequence of feature-based decision rules.



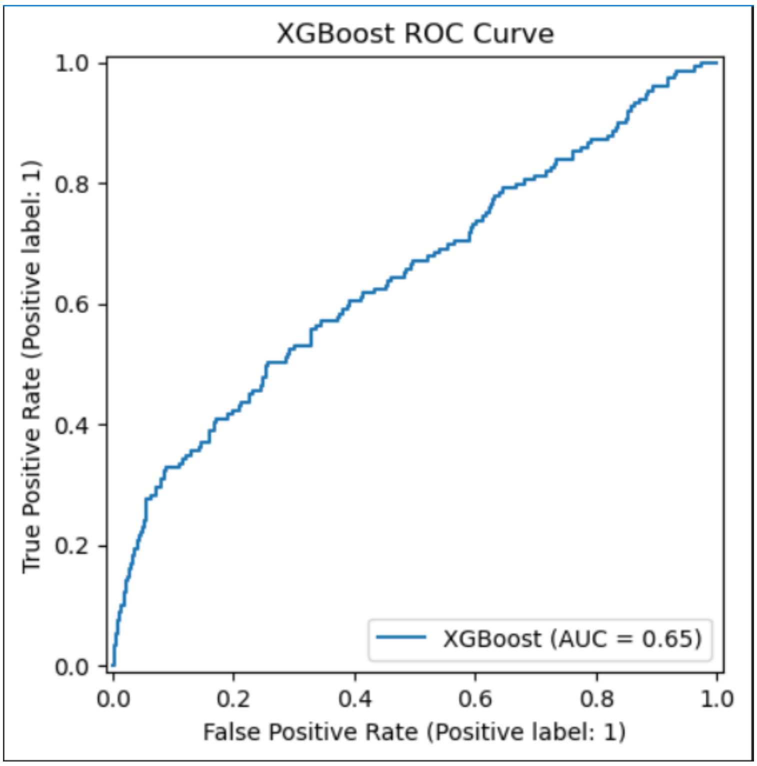
Random Forest

Random Forest was implemented as an ensemble learning technique that combines multiple decision trees. By aggregating the predictions of several trees, Random Forest can capture complex relationships while reducing the limitations of an individual decision tree.



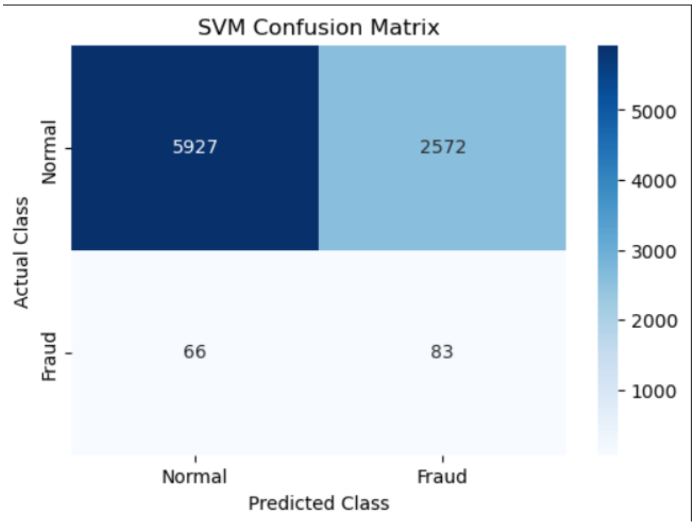
XGBoost

XGBoost was implemented as a gradient boosting model for structured financial transaction data. It builds multiple sequential decision trees and attempts to improve the errors made by previous trees.



Support Vector Machine

Support Vector Machine was implemented as an additional classification technique. It attempts to identify an optimal decision boundary for separating fraudulent and legitimate transactions.



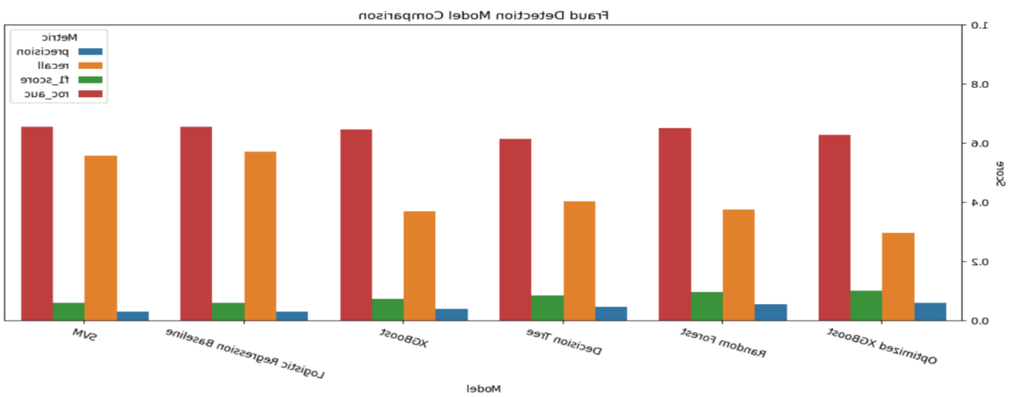
9. MODEL EVALUATION AND COMPARISON

Since financial fraud detection is an imbalanced classification problem, several evaluation metrics were considered instead of relying only on accuracy.

The models were evaluated using **accuracy, precision, recall, F1-score and ROC-AUC**

Random Forest achieved an accuracy of approximately **87.84%**, while Logistic Regression achieved a higher recall of approximately **57.05%**. This difference demonstrates the trade-off between correctly identifying fraudulent transactions and limiting incorrect fraud predictions.

Therefore, model selection should consider the business objective and the relative importance of false positives and false negatives rather than relying on accuracy alone.

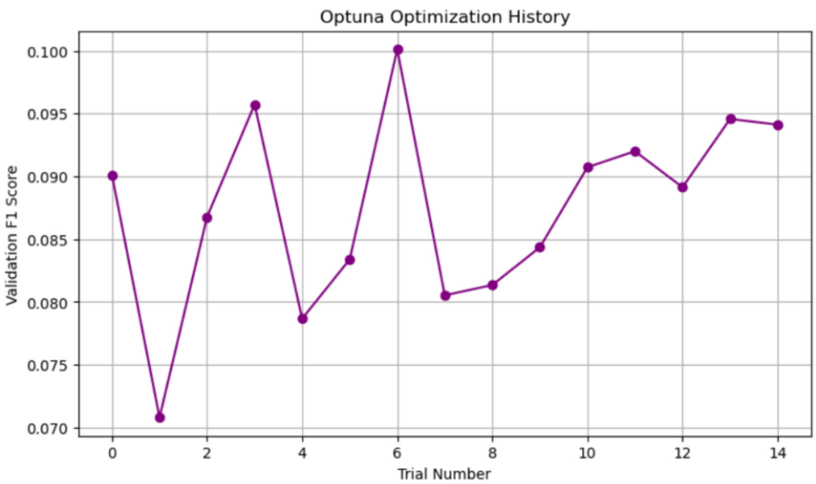


10. HYPERPARAMETER OPTIMIZATION

After developing the baseline models, hyperparameter optimization was performed using **Optuna**.

The purpose of optimization was to search for model configurations that could provide improved predictive performance compared with default parameters. Different parameter combinations were evaluated according to the selected optimization objective.

The optimized model was subsequently evaluated using the same test metrics to determine whether optimization produced a meaningful improvement.



11. ANOMALY DETECTION

Supervised classification models learn from previously labeled fraud examples. However, newly emerging fraud patterns may not always resemble the fraudulent transactions present in the training data.

To address this limitation, anomaly detection techniques were explored as complementary approaches.

Isolation Forest

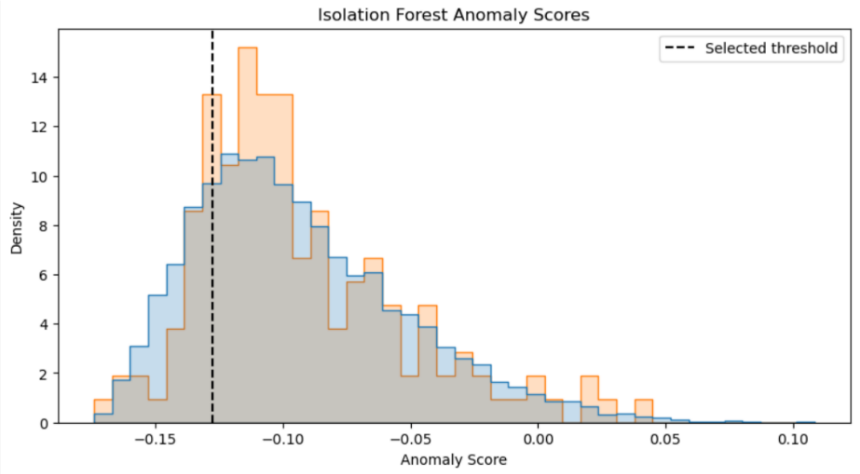
Isolation Forest identifies observations that are significantly different from the majority of observations.

The obtained results were:

- Accuracy: **50.75%**
- Precision: **63.44%**
- Recall: **3.53%**

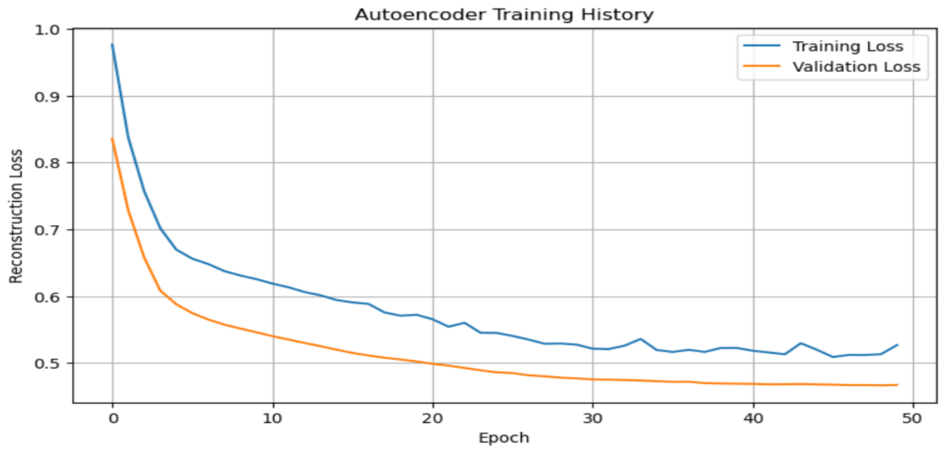
- F1-score: **6.70%**

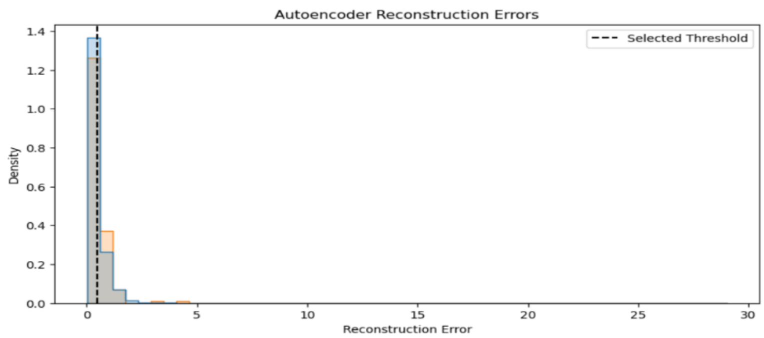
The results indicate that Isolation Forest produced relatively high precision but very low recall. Therefore, it was not considered suitable as a standalone fraud classifier. However, its anomaly information can be useful as an additional signal in a hybrid fraud detection system.



Autoencoder

An Autoencoder-based approach was also explored to identify transactions with unusual reconstruction behaviour.





12. HYBRID RISK SCORING AND EXPLAINABLE AI

To improve the fraud detection framework, supervised prediction and anomaly detection were considered together.

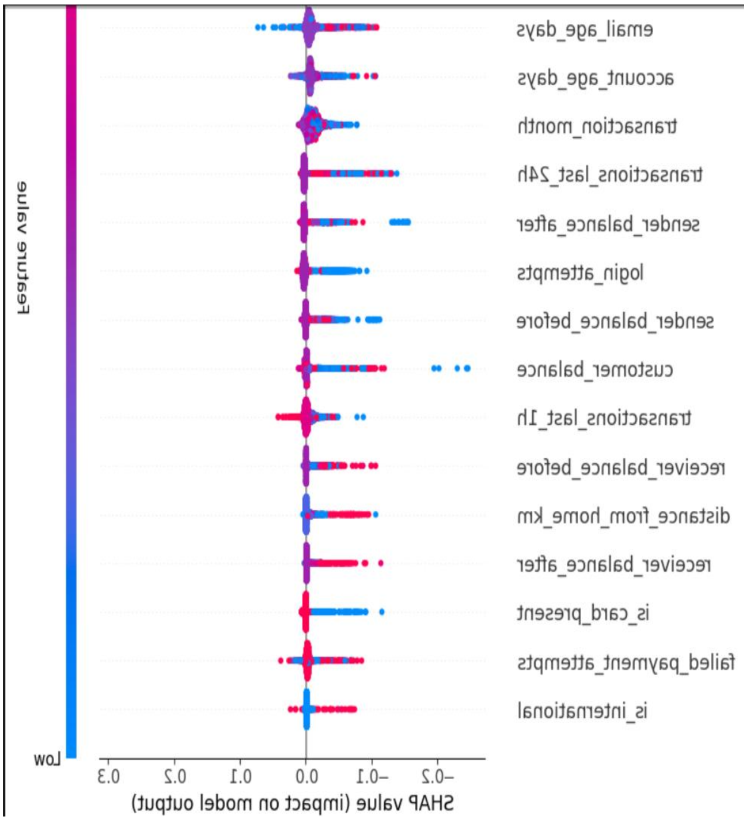
The supervised model provides information about the probability that a transaction is fraudulent, while the anomaly detection component provides information about how unusual the transaction is compared with normal transaction behaviour.

These signals can be combined to produce a **transaction risk score**, which can then be used to categorize transactions into different risk levels.

Explainable AI

SHAP was incorporated to improve the interpretability of the model. SHAP identifies the contribution of individual features to a prediction and helps explain why a particular transaction received a high or low fraud probability.

This is particularly valuable in financial applications because investigators need meaningful reasons behind suspicious transaction alert.



13. REAL-TIME FRAUD DETECTION

The project was extended from offline model evaluation toward a real-time fraud detection architecture.

Kafka was used as the streaming component for handling incoming transaction events. The basic workflow is:

Transaction → Kafka Producer → Kafka Topic → Consumer → Preprocessing → ML Model → Risk Score → Alert

When a new transaction enters the system, it can be processed automatically by the fraud detection pipeline. The resulting fraud probability and risk score can then be used to determine whether the transaction requires further investigation.

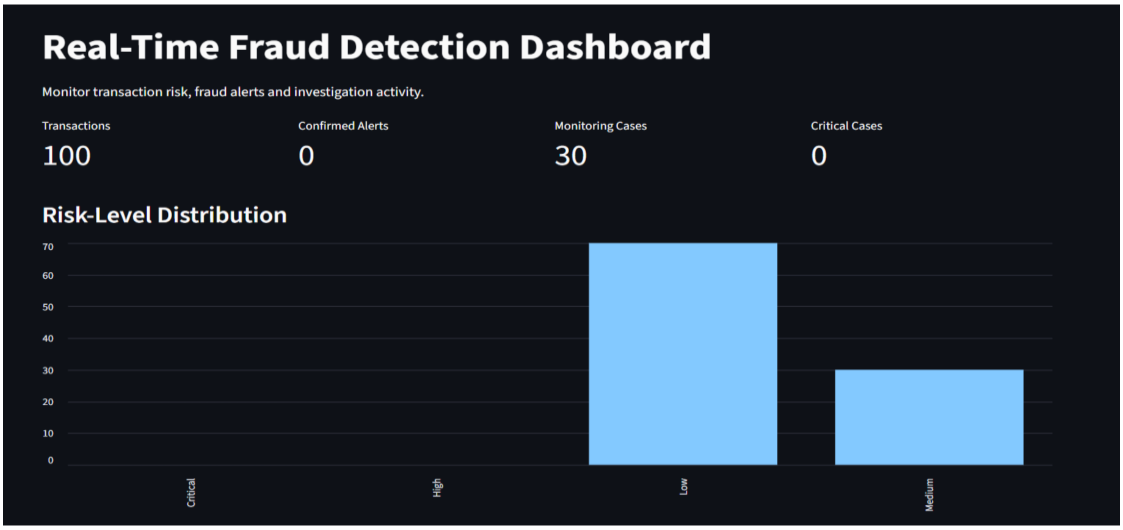
This architecture demonstrates how the machine learning model can be integrated into a real-time financial monitoring environment.

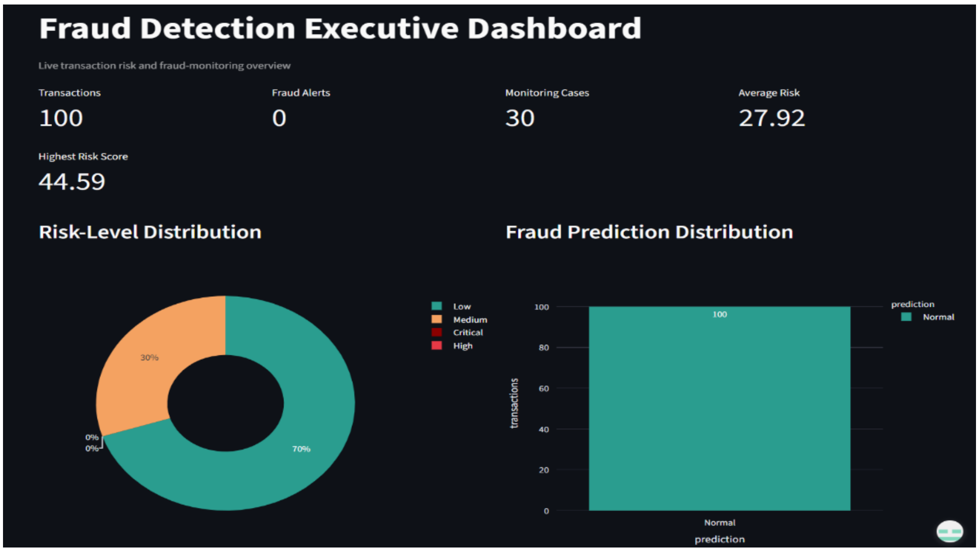
14. STREAMLIT FRAUD MONITORING DASHBOARD

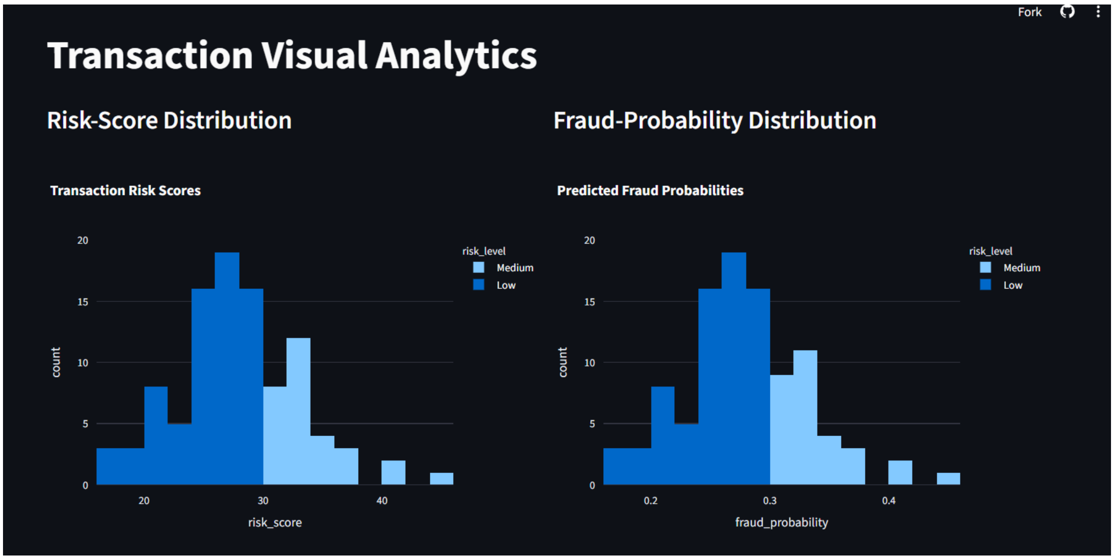
A Stream lit dashboard was developed to provide an interactive interface for monitoring fraud detection results.

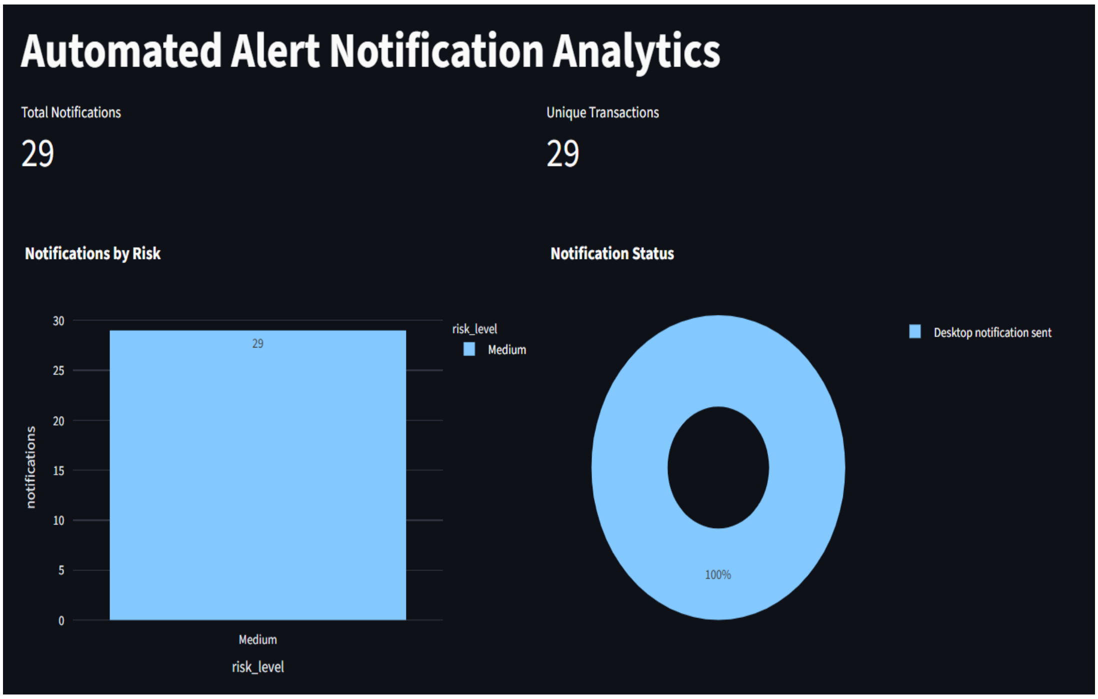
The dashboard presents relevant transaction information, fraud predictions, risk scores and model outputs in an accessible format. It can also be used to support transaction-level investigation and visualize important fraud detection statistics.

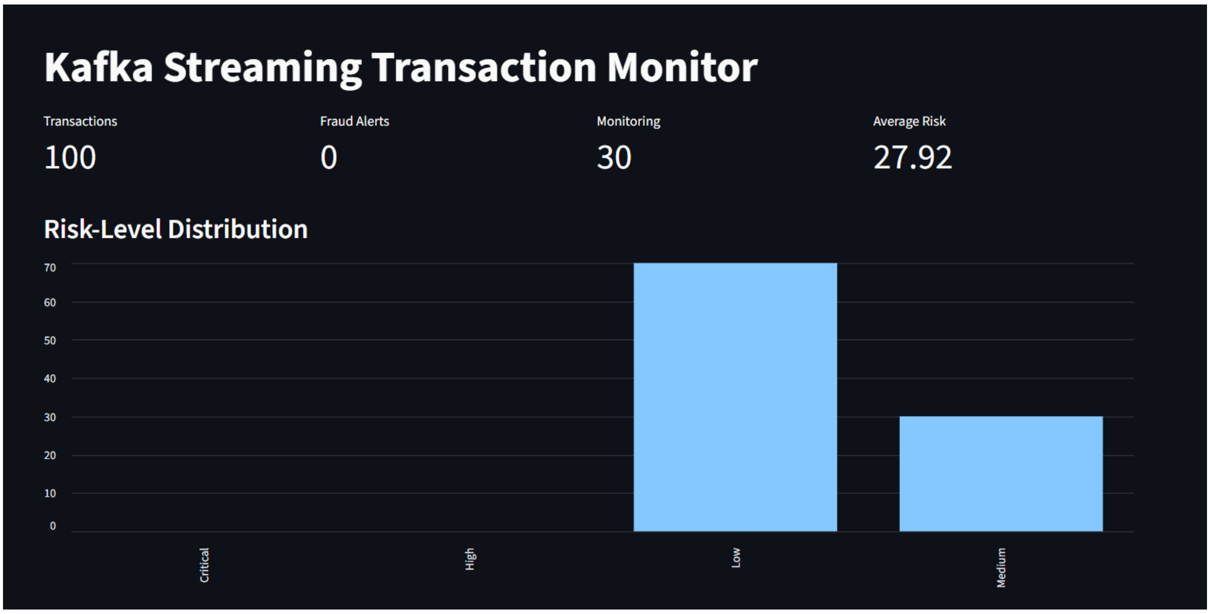
The dashboard provides a practical user interface over the underlying machine learning pipeline and demonstrates how the developed system could be used by analysts or fraud investigators.




Fraud Detection Executive Dashboard
Live transaction risk and fraud-monitoring overview
Transactions
100
Fraud Alerts
0
Monitoring Cases
30
Average Risk
27.92
Highest Risk Score
44.59
Risk-Level Distribution
Fraud Prediction Distribution
Low
Medium
Critical
High
30%
70%
0%
0%
prediction
Normal
100
transactions
Normal
prediction


Fork
Transaction Visual Analytics
Risk-Score Distribution
Fraud-Probability Distribution
Transaction Risk Scores
Predicted Fraud Probabilities
risk_level
Medium
Low
count
risk_score
fraud_probability


Automated Alert Notification Analytics
Total Notifications
29
Unique Transactions
29
Notifications by Risk
Notification Status
risk_level
Medium
29
notifications
Medium
risk_level
Desktop notification sent
100%



Fraud Detection Dashboard Stream Lit:

https://financial-fraud-detection-dashboard-hrcfsbfcqkub96nsd7bflu.streamlit.app/

15. CONCLUSION

This project developed a comprehensive machine learning-based framework for financial fraud detection. Multiple financial datasets were integrated and processed to create a unified transaction dataset. Exploratory analysis was performed to identify transaction and behavioural characteristics associated with fraudulent activity.

The severe class imbalance in the training data was addressed using SMOTE, after which multiple machine learning models were trained and evaluated. Logistic Regression, Decision Tree, Random Forest, XGBoost and SVM provided different trade-offs between accuracy, precision and recall.

The project was further extended using hyperparameter optimization and anomaly detection. Isolation Forest demonstrated that anomaly detection can provide useful information about unusual transactions, although its low recall indicates that it should be used as a complementary technique rather than as the sole fraud classifier.

Hybrid risk scoring and SHAP explainability were introduced to improve decision-making and model transparency. Finally, Kafka-based streaming and a Streamlit dashboard were used to demonstrate the transition from an offline machine learning model to a real-time fraud monitoring system.

Overall, the project demonstrates an end-to-end approach to financial fraud detection combining data analysis, machine learning, class-imbalance handling, anomaly detection, explainable AI and real-time processing.